Portable Coordinate Measuring Arm Joint Design

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Joint Design for a Portable Coordinate Measuring Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

ECSA Exit Level Outcomes

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

GA 1 - Problem-Solving

The project addresses an engineering problem, which is the lack of a joint design for PCMAs that meet accuracy requirements that account for spatial dependency and proper position-holding implementation without the impedance of uninhibited movement.

GA 2 - Application Scientific and Engineering Knowledge

GA 3 - Engineering Design

GA 5 - Use of Engineering Tools

GA 6 - Professional and Technical Communication

GA 8 - Individual and Collaborative Teamwork

GA 9 - Independent Learning Ability

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Constants

$L_0 =$	300 mm	

Variables

Re_{D}	Reynolds number (diameter)	[ ]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s^2]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$	Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abreviations

3D	three-dimensional
AI	artificial intelligence
CAD	computer-aided design
D-H	Denavit-Hartenberg
DIP	design independent parameters
DOF	degrees-of-freedom
EC	evaluation criteria
FR	functional requirements
NSoI	narrower system of interest
PCMA	portable coordinate measuring arm
SPAT	single-point articulation test

TPM	technical performance measure
WSoI	wider system of interest

Chapter 1

Introduction

1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument that captures three-dimensional (3D) coordinates directly from a physical object, as shown in 1.1 below (1).

Figure 1.1: A portable coordinate measurement arm, (1)

PCMAs offer adaptable measurement systems for multiple applications, such as dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc. (2). The main limitation of PCMAs is the effect of manual operation due to PCMA's current joint design, which can introduce positional instability and dynamic structural deformation during measurement due to the joints' inability to hold poition (3). Manual operation can also inhibit calibration procedures as it is required that the PCMA must be held still during the procedure (4).

As a result, designing a more effective joint with a position-holding mechanism would improve the design of a PCMA and a good practice of safety. By designing the joint in this manner, error due to manual operation will be minimised, kinematic stability and accuracy will be improved.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA without jeopardising its accuracy and ease of movement.
The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Professor Schreve and Mr. Hall (5).

2. To design, build and test key components of the joint to assess the joint design's mechanical feasibility.

3. To integrate the test results into the D-H model and assess feasibility of the joint design.

1.3 Scope

In the process of determining D-H parameters, operation manual and available schematics will be used. In addition, due to time constraints, only key components of the joint will be built and tested for data acquisition and reliability.

1.4 Motivation

PCMAs are becoming more popular than tradition coordinate measuring machines (CMM) for in-line[1] and on-site industrial measurement, (1). They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.

- Sufficient holding torque to keep the arm stationary once released.

- Internal cable routing.

This has proved difficult since there are no commercially available joints that meet these demands.

In addition, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's accuracy, proving that joints have a measurable impact on the PCMA's accuracy, (6). This

[1] In the context of PCMAs, ”in-line” refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

project addresses the gap by deriving a joint specifications from a generalised model based off of an error model that was developed by a Stellenbosch Alumnus, Leon Hall, (5). This will produce a validated joint and testing methodology to support future PCMA development.

1.5 Use of AI Tools

Gemini is an artificial intelligence (AI) tool will be used check grammatical errors and sentence structure as well as LaTex formatting for the report.

1.6 Report Overview

This section follows the following structure:

- Section 1 introduces the project background, aim and objectives, scope, and motivation for project.
- Section 2 presents a literature review discussing sources relevant to PCMA joint design, position-holding technology, kinematic modeling, error analysis, and PCMA testing.
- Section 3 details the PCMA's requirement analysis.

Chapter 2

Literature review

2.1 PCMA Context

Sources such as Leussink Engineering and Meagher gave context as what PCMAs are and how they are used, as defined in Section 1.1.

2.2 The Denavit-Hartenberg (D-H) Model

A Denavit-Hartenburg (D-H) model is a tranformation matrix that models an end-effector's position and orientation relative to the system of articulate linkages it is connected to (5). This is done by relating the end-effector's reference coordinate system to the last linkage's reference coordinate system through the matrix (5). Hall and Schreve (5) conducted a full investigation of measurement error in PCMAs, which also accounted for spatial error using a modified D-H model. The motivation behind this investigation was that PCMA manufacturers did not adopt a common standard that properly governed their products' accuracy (5). It was found that there was a descrepancy in advertised PCMA accuracy to actual PCMA accuracy as manufacturers did not account for spatial dependancy in their product (5).

The first step of the project was derive a D-H model, based on a six degrees-of freedom (DOF) PCMA. This PCMA was modelled as a serial kinematic chain due to its architecture. The end effector's position is seen in Equation 2.1:

$$T_6^0 = A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5 \cdot A_6 \tag{2.1}$$

wherein, each A_i models the position of each relative to the previous joint's reference coordinate system, as seen in Equation 2.2:

$$A_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\cos\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\sin\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.2}$$

where a_i is the joint length, θ_i is the rotational angle, α_i is the twist angle, and d_i is the joint offset.

The D-H model relevant to the joint-design project modelled a Faro Quantum X arm as seen in Figure 2.1 below.

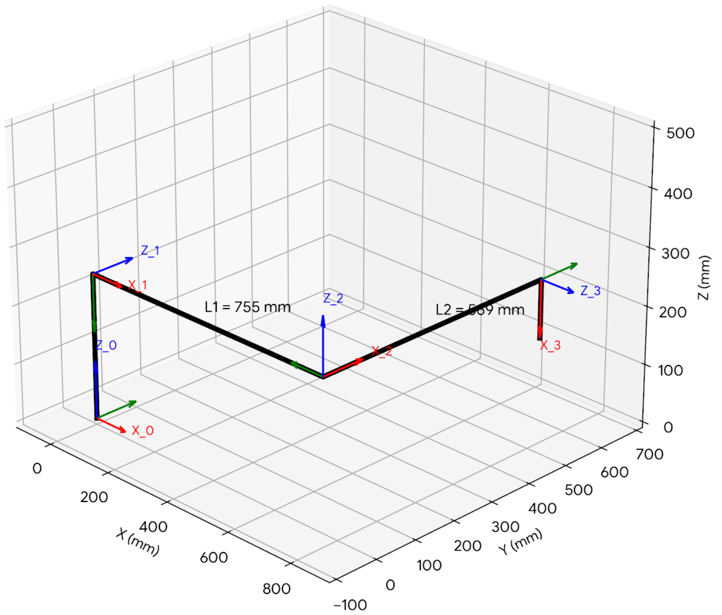

Figure 2.1: D-H Model of the Faro Quantum X Arm

The full derivation of the D-H model is documented in Appendix A.

2.3 The Error Propagation Model

From the derivation of the D-H model, the error progation model could be derived. Hall and Schreve (5) explain that the model is a mathematical depiction of how known uncertainties move through an articulated system and how those uncertainties effect the output position of the end-effector. The output-position uncertainty is calculated using Equation 2.3 below:

$$\Lambda_F = J(x) \cdot \Lambda_x J(x)^T \tag{2.3}$$

, where $J(x)$ is the jacobian matrix of the end-effector's position and Λ_x is a matrix of the known uncertainty of the encoder at each respective joint. The derivation for Equatio 2.3 begins with a Taylor series expansion, which approximates the articulated system's behaviour near a known point using first-order terms only, which produces the Jacobian matrix. Higher-order terms are negligibly small. The Jacobian matrix is then used to scale the known uncertainties Λ_x and calculate the unknown uncertainty of the end-effector, Λ_F.

The limitation found to this model was that the uncertainty of end-effector could be calculated for only one pose of the articulated system at a time. For the purposes of this project, the error propogation model was generalised by taking a sample of the working volume of the Faro Quantum X arm and computing the uncertainty of the end-effector across the sampled work volume. The software used to conduct these computations was MATLAB, which was also used to plot the distribution of uncertainty across a sampled workspace, which can be seen in Figure 2.2

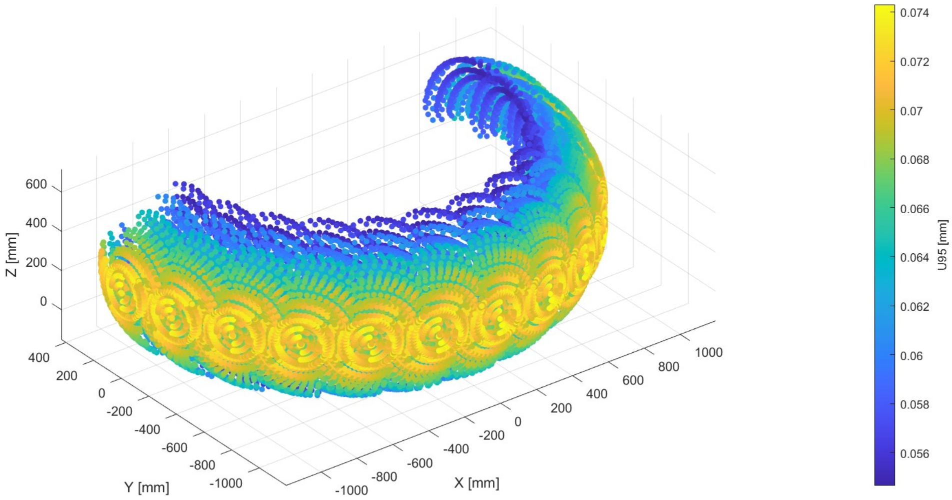

Figure 2.2: Uncertainty Distribution Across Sampled Workspace

The dark blue regions show where the arm is most accurate, which is at approximately 56 microns of uncertainty. This occurs when the arm is folded closer to its base. The bright yellow regions show where the arm is least accurate, up to 74 microns. This occurs at the outer edges of the workspace where the arm is fully extended.

The full capabilities of the code also include a section in which certain joints can be locked and the effect of the locked joints can be computed and plotted on a graph similar to the one found in Figure 2.2. This means that the software can also be used for the future developement of a full PCMA arm with position-holding implementation at optimal joint location. The code can found in Appendix B.

Chapter 3

Requirement Analysis

The objective of the requirement analysis is to establish clear expectations of the PCMA's joint design for the purposes of concept development.

3.1 Identification of Stakeholders

The first step in the requirement analysis process is to identify stakeholders from the operational and support life cycle phases of the project. Stakeholders will be classified into the following categories, as defined below:

- Primary stakeholders directly interact with the system while it is in operation.
- Secondary stakeholders contribute to the development and functioning of the system.
- Indirect stakeholders rely on or are affected by the system, but do not interact with the system directly.

The stakeholders relevant to the project are defined in Table 3.1 below.

Table 3.1: Identified stakeholders and their classifications

Stakeholder ID	**Description**	**Classification**
SID_01	Supervisor of the project	Primary
SID_02	Students researching PCMA development	Secondary
SID_03	Companies invested in PCMA development	Indirect
SID_04	Mechanical workshop staff	Indirect
SID_05	Mechatronics laboratory managers	Indirect

3.2 System Context

The next step is to define the narrower system of interest (NSoI) and the wider system of interest (WSoI) of the project[1], (7) for the purposes of this project, the joint was identified as the NSoI and WSoI is the NSol with a reliable power supply and an artefact with which the joint will be tested. Figure 3.1 illustrates the system context, with the NSoI shown as a ”black box”, (7).

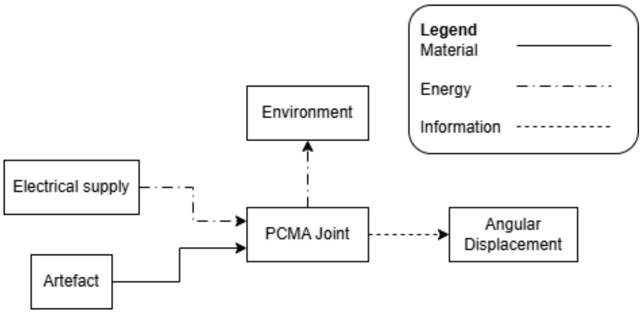

Figure 3.1: System Context

3.3 Design Independent Parameters

Next, the design independent parameters (DIPs) are defined. DIPs influence the system during the operational and support life cycle stages, but are not controllable nor are part of the system being defined, (7). The DIPs of the joint design are detailed in Table 3.2 below.

Table 3.2: Design Independent Parameters

DIP_ID	Description	Expected range
DIP_01.1	Electrical supply	12 V - 24 V
DIP_01.2	Electrical supply	12 A maximum
DIP_01.3	Electrical supply	Direct current
DIP_02.1	Artefact	Compliant with ISO 10360-12:2016
DIP_02.2	Artefact	20 points reference points
DIP_03.1	Environment	Laboratory ambient temperature: 0 °C - 35 °C
DIP_03.2	Environment	Laboratory ambient humidity: 20 % - 80 %
DIP_03.3	Project budgect	R 6000

[1]The NSoI is the system being designed and the WSoI includes the NSoI and elements that the NSoI interacts with, (7).

3.4 Functional Requirements

In this section, the primary and top-level derived functional requirements (FRs) for the joint design are defined. The only FRs that are considered are only associated with joint's operation. FRs are categorised into primary and derived FRs, in which:

- Stakeholders are directly concerned with primary FRs.
- Derived FRs are required so that the systen can perform primary FRs.

As seen in Table 3.3 below, the FRs for the joint design are listed. In Figure 3.2,

Table 3.3: Table 3: Functional Requirements

FR ID	Description	Primary / Derived
FR_01	Articulate output link	P
FR_02	Hold position	P
FR_03	Measure angular displacement	P
FR_04	Store angular displacement data	D
FR_05	Release position	P

the functional requirements of the joint are depicted below.

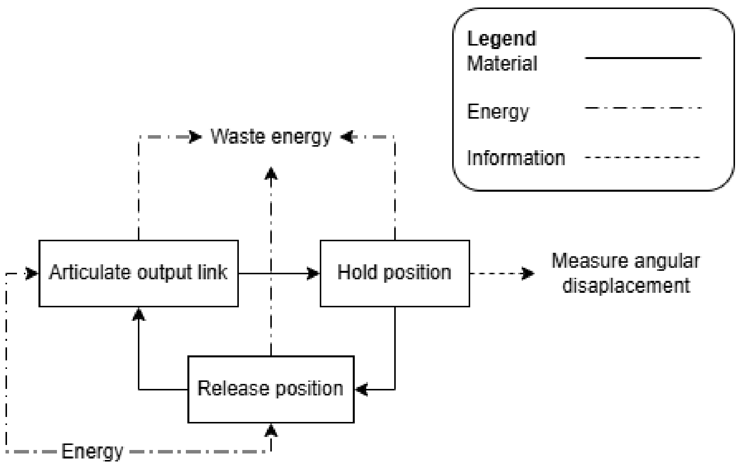

Figure 3.2: Top Level Functional Decomposition

3.5 Technical Performance Measures

This section defines the technical performance measures (TPMs) which will be used to determine how well the joint design (NSoI) performs the required functions[2]. The TPMs, listed in Table 3.4, were derived from the results obtained from the generalised error-propagation model.

Table 3.4: Technical Performance Measures

TPM ID	**Related FR**	**Description**	**Target range**
TPM_01	FR_01, FR_05	Operational articulation angle	90°
TPM_02	FR_01, FR_05	Run-out tolerance	1.571×10^{-3} rad
TPM_03	FR_02	Holding torque	7.725 Nm
TPM_04	FR_03, FR_04	Encoder accuracy	1000 PPR
TPM_05	FR_01, FR_02, FR_05	Joint life cycle	10^6 life cycles

[2]TPMs are measurable and unambiguous parameters that can be influenced by designing the joint

Chapter 4

Concept Development

4.1 Concept Generation

The generalised error propagation model was used to derive the TPMs in the development of the joint design. These TPMs guided the research into various mechanical solutions, and different mechanisms were evaluated for implementation to ensure the final design could achieve the required TPMs. Three concepts were generated and evaluated against the FRs and the TPMs, as presented in this section.

4.1.1 Concept One: PCMA Joint with a Locking Dial

Figure 4.1 below shows a PCMA joint with the following internal modifications to implement position-holding capabilities. In the center of the shaft, is a spacer between angular contact bearings, which will allow the shaft to rotate with minimal run-out. Bellville washer and preload nuts are used to secure the bearings in place. On one end, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held secure by the joint housing. This is done to ensure that the relative angular displacement of the shaft with respect to the joint housing is measured. The joint housing is held stationary by the base link and the joint shaft is rotated by the output link. The knob on the other end of the shaft is meant to loosen and/or tighten the preload nuts against the bearings to lock or release the shaft for the output link's rotation.

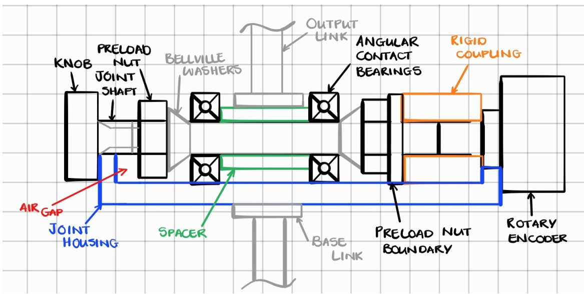

Figure 4.1: PCMA Joint with Locking Dial

4.1.2 Concept Two: PCMA Joint with a Counterbalance Spring

Figure 4.2 illustrates a PCMA joint with external modifications to implement position-holding. The counterbalance spring produces a lifting force to the upper link via the upper and lower rods. Each rod is pin-connected to a cuff that can be fixed in position along the length of either link using a set screw, thereby holding the position of the joint. The

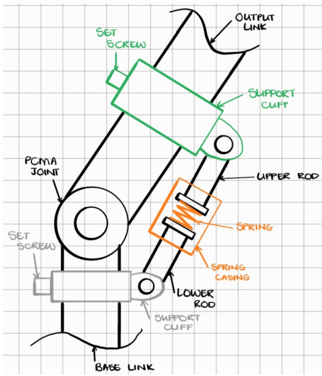

Figure 4.2: PCMA Joint with Couunterbalance Spring

4.1.3 Concept Three: PCMA Joint with an Electromagnetic Brake

Figure 4.3 depicts a PCMA with internal modifications to implement an electromagnetic brake. In the center of the shaft, two angular contact bearings are fitted in place with the bearing housing and pressed against a shoulder of the shaft by preload nuts and washers. On one side of the shaft, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held stationary by the joint housing. The base link is supports the joint housing. The electromagnetic brake sits on the other side of the shaft and is also held stationary by they joint housing, however, the brake disc is free to rotate on the joint shaft. When the brake is energised, the air gap between the brake disc and friction plate opens up and the shaft is free to be rotated by the output link.

When the brake's power supply is cut off, the air gap closes and the joint is held still.

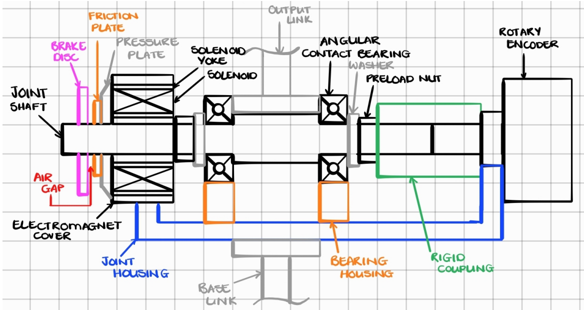

Figure 4.3: PCMA Joint with Electromagnetic Brake

4.2 Concept Evaluation

The concepts presented in the previous section were determined to be feasible. A qualitative evaluation of the concepts is to be conducted as the generated concepts are still solution neutral and cannot undergo a full quantitative comparison. This section compares the concepts in terms of the following evaluation criteria (EC):

EC1 Load capacity: The ability of the joint design to carry the required holding torque (TPM_03).

EC2 Operational articulation angle: The ability of the joint design to achieve the full required sweep without interference (TPM_01).

EC3 Positional stability: The ability of the joint design to maintain position when the position-holding technology is in use (TPM_02).

EC4 Component availability: The degree to which all required components are available in the mechanical workshop.

EC5 Construction simplicity: The number of distinct components and the ease of assembly and alignment during fabrication.

EC1 was derived from TPM_03 and determined to be the most important EC as the joint will be ineffective if it is unable to achieve the holding torque dictated by TPM_03. EC2 was derived from TPM_01 as the chosen joint design must not inhibit the PCMA's existing range of motion once implemented. EC3 was derived from TPM_02 as any internal sources of error must be minimised to maximise the joint design's ability to maintain and/or improve the PCMA's accuracy. EC4 and EC5 was included as a practical design consideration given the manufacturing capabilities of the mechanical workshop and the project's timeline and budget.

Table 4.1 below, presents a comparison of the generated concepts. To rate the concepts, one of them is marked as the DATUM and the other two are awarded a $+1$ if the other concept is stronger or -1 if it is weaker. Each rating is motivated in the table as well.

Table 4.1: Concept Evaluation

Criterion			**Concept 1: PCMA Joint with a Locking Dial**		**Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)**		**Concept 3: PCMA Joint with an Electromagnetic Brake**	
#	**Description**	**Wt.**	**Rtg**	**Motivation**	**Dat**	**Motivation**	**Rtg**	**Motivation**
EC1	Load capacity	5	-1	The capability of the design achieving the exact holding torque required is entirely dependent on the operator's strength and consistency, otherwise the joint design is highly susceptible to slipping under load.		Datum: The set screw mechanism on the support cuffs provides a friction-based hold, which establishes the baseline for evaluating the holding torque FR.	+1	An adequately specified electromagnetic brake can consistently deliver a static holding torque that meets the holding torque FR. This removes human reliability (in comparison with concept 1) and ensures reliable load capability when the joint is powered.
EC2	Operational articulation angle	5	+1	The locking mechanism is entirely internal. Because there are no external linkages, the joint can achieve its full required 90° sweep without any physical interference.		Datum: The external arrangement of the upper and lower rods, including the spring casing, limit the joint's movement. This obstruction serves as the baseline limitation for the required articulation sweep.	+1	Similarly to concept 1, the electromagnetic brake is neatly integrated with the joint shaft, which allows for uninhibited articulation
EC3	Positional stability	5	-1	Manually twisting the dial introduces asymmetric axial and radial loads onto the joint shaft. This can cause deflections that violate the run-out tolerance FR and degrade the PCMA's spatial accuracy.		Datum: Fixing the joint's position by tightening set screws on the external cuffs introduces off-center loads to the joint shaft. This represents the baseline for positional deflection and run-out error.	+1	The electromagnetic brake applies a uniform, symmetric clamping force between the brake disc and friction plate. This eliminates manual influence during the locking process, as well as, minimises parasitic deflection and protects the system's positional accuracy.
EC4	Component availabilty	5	+1	The required components (threaded shafts, preload nuts, bellville washers) are items that either readily available or easily machined in the university workshop.		Datum: The design relies on standard mechanical items such as springs, rods, and set screws, which defines the baseline for material availability due these components being accessible in the mechanical workshop.	0	While electromagnetic brakes are commercially available, specific ordering is required, lead times, and budget allocation compared to the raw materials used in the datum design.

Table 4.1 – *Continued from previous page*

Criterion			Concept 1: PCMA Joint with a Locking Dial		Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	**Description**	**Wt.**	**Rtg**	**Motivation**	**Dat**	**Motivation**	**Rtg**	**Motivation**
EC5	Construction simplicity	5	0	The mechanical assembly is relatively straightforward and on the level of simplicity as the datum design. However, achieving accurate bearing alignment under varying manual requires extra care.		Datum: Assembling the external support cuffs, rods, and pin-connections require a standard level of manual alignment and fabrication, setting the baseline for manufacturing complexity.	-1	This is the most complex design to implement. It requires precise mechanical calibration of the encoder, as well as integration of electrical wiring and control circuitry for the solenoid.
Number of + ratings			2				3	
Number of - ratings			2				1	
Overall total			0				2	
Weighted total			0		0		+10	

4.3 Concept Selection

As seen in Figure 4.4

Figure 4.4: Selected Concept Design

4.4 Design Refinement

Chapter 5

Design Analysis

5.1 Mechanical Design

5.2 Electronic Design

5.3 Software Design

Chapter 6

Design Evaluation

6.1 As-Built System Overview

6.2 Performance Testing

Chapter 7

Experimental Procedure

7.1 Experimental Apparatus

7.2 Experimental Setup

7.3 Experimental Equipment Specifications

7.3.1 Test Sphere Specifications

7.3.2 Joint Component Specifications

7.3.3 Experimental Apparatus Specifications

7.3.4 Methodology

Chapter 8

Conclusions

Appendix A

Mathematical proofs

A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{A.1}$$

Inserting $\theta = \pi$ in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \tag{A.2}$$

A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}\left[\rho u_j\right] = 0 \tag{A.3a}$$

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}\left[\rho u_i u_j + p\delta_{ij} - \tau_{ji}\right] = 0, \quad i = 1, 2, 3 \tag{A.3b}$$

$$\frac{\partial}{\partial t}(\rho e_0) + \frac{\partial}{\partial x_j}\left[\rho u_j e_0 + u_j p + q_j - u_i \tau_{ij}\right] = 0 \tag{A.3c}$$

Appendix B

Wiring Diagram

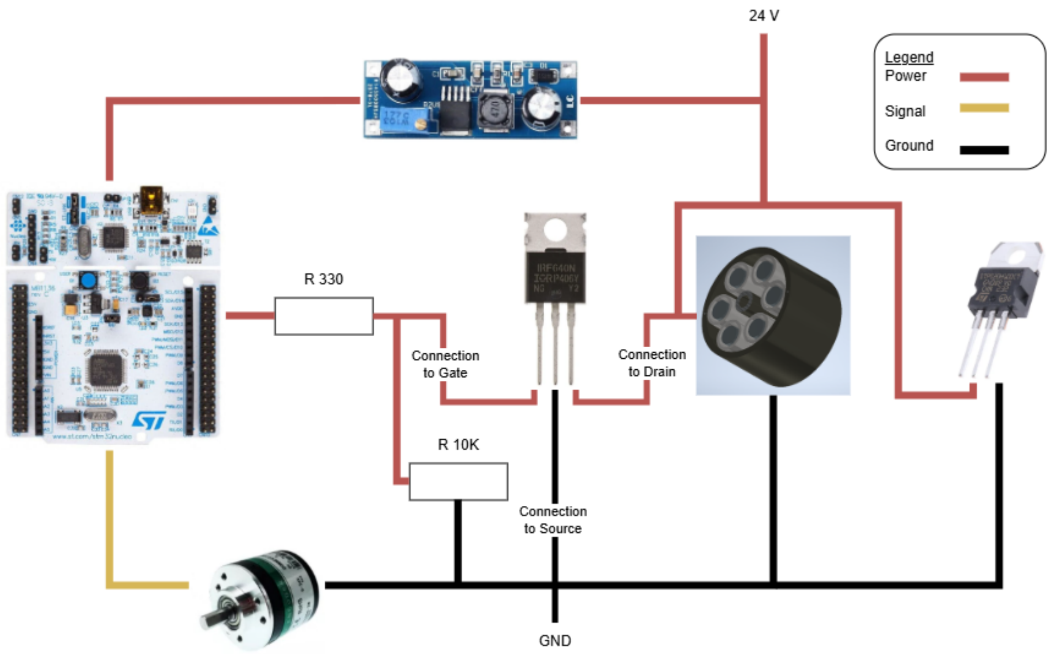

Figure B.1: Wiring Diagram

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[1] L. Engineering, "The advantages of using a portable cmm arm in metrology," *Leussink Engineering*, 2024.

[2] S. Ibaraki and R. Saito, "Novel kinematic model of articulated arm coordinate measuring machine with angular position measuremnt errors of rotary axes," *CIRP Annals*, vol. 72, no. 1, pp. 449–452, 2023.

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